

# PAPER\_1664 — Vicsek Flocking Density = $\rho_{\text{flock}} = \beta_i \times \Phi_{\text{res}} = 0.6029 \times 0.84 = 0.506$

**Author:** Daniel T. Murphy **Framework:** UQFF (Unified Quantum Field Framework) — Star-Magic v5.27+ **Tier:** Active Matter **Date:** June 18, 2026 **Status:** CLOSED — 0.087% closure (PAPER\_134x broader corpus)

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## Observation

PAPER\_1356 (PAPER\_134x condensed matter / quantum sector) gives:

$$\rho_{\text{flock}} = \beta_i \times \Phi_{\text{res}} = 0.6029 \times 0.84 = 0.506$$

Status: **0.087%**.

## UQFF Closed Identity

$$\rho_{\text{flock}} = \beta_i \times \Phi_{\text{res}} = 0.6029 \times 0.84 = 0.506 \quad 0.087\%$$

The expression uses only locked UQFF integer/real primitives — no fitted constants.

## NOT REPLACEMENT

Experimental active matter measures this constant directly. UQFF supplies a structural derivation from the integer primitives.

## Reference

- Source: PAPER\_1356
- Related: PAPER\_1646-1655 (tier-23 broader corpus); PAPER\_1636-1645 (tier-22 SM)
- Calculator dispatch: `calculate_paradox({"paradox": "flocking_rho_0_506"})`

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